

FACULTY OF BUSINESS AND IT
MBAI 5600G - Applied Integrative Analytics Capstone Project
Milestone 2 Report

**AI-Based Student Dropout Early Warning System for
Higher Education: Reproduction and Cross-Dataset
Generalization on UCI and OULAD**

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1. Executive Summary

The purpose of the capstone was to develop a predictive model for identifying students who drop out of higher education. The current version of the capstone has evolved into a replication-and-extension study that uses machine learning algorithms to predict dropout from a higher education perspective. Instead of using a singular dataset for a typical classification problem, the new design will initially replicate the explainable machine learning approach used by Islam et al. (2025) [1], which used the Explainable Machine Learning Workflow to classify data from the UCI Predict Students' Dropout and Academic Success dataset. Following that, the same types of models will be evaluated to determine their ability to generalize to the Open University Learning Analytics Dataset (OULAD). There have been several studies showing that tree-based ensemble models can perform very well at predicting student retention within a given dataset; however, there has also been an increasing interest in developing models based upon Explainable AI. As such, there is relatively little research regarding the generalizability of these models when transferring them to other datasets, nor concerning whether or not the importance assigned to each variable would remain stable if they were transferred to a different environment. In order to fill this gap, OULAD will become the main focus of the analysis while UCI will continue to serve as a benchmark and reproduction dataset.

2. Literature Review

This literature review uses a total of 27 scholarly articles as its basis for reference; there are 15 references cited in this review which have been written in the years 2024 through 2026. Instead of organizing this literature review by individual article (as would be typical), this review will organize based upon themes of research in order to align with the milestone direction. There are clearly three main paths that student dropout/academic risk prediction in post-secondary education have followed in their development. These include: using structured administrative/admissions data, behavioral learning analytics data collected from Learning Management Systems/Virtual Learning Environments, and developing methods to provide explanatory support to decision-making regarding use of predictive models [1]-[27].

2.1 Thematic Overview

The review has a thematic organization so as to show relationships between benchmark work in structured institutional data, learning analytics that utilize behavioral traces; explainable AI studies; and the growing necessity of cross-dataset validation. The thematic arrangement is suitable for this capstone since it is now an objective of the project both to replicate earlier studies and to assess portability and explanation stability among education data sets.

Theme	Representative Studies	Main Takeaway
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Administrative and admissions data models	[2]-[7], [13], [21], [22]	Tree-based ensembles and structured institutional variables are commonly used and often produce strong multi-class or binary retention results.
Learning analytics and behavioural data	[8], [9], [17], [22], [23]	Temporal LMS and VLE activity can improve prediction, but requires heavier preprocessing, aggregation, and leakage control.
Explainability and decision support	[1], [10], [18], [20], [23], [24]	Recent work increasingly argues that predictive accuracy alone is not enough; interpretability and intervention usefulness matter.
Reviews, trends, and determinants	[12], [15], [19], [23], [24]	The field still faces gaps in generalisability, metric selection, ethics, and the connection between analytics and actual institutional action.

2.2 Administrative and Admissions-Based Prediction Models

In addition to the many studies on dropouts in higher education that utilize structured institutional data (such as demographics, admissions data, academic history, first semester grades and financial metrics) structured institutional data-based studies have several advantages. They are very attractive due to the fact that they typically consist of "clean" data that is scalable and based upon data that the university currently stores in their operational systems. Realinho et al., presented the UCI multiclass dataset that was utilized within this project and indicated how it could be used to create benchmarks for student outcomes predictions [13]. More recently, other researchers have followed this line of research utilizing both private and public institutional data. For example, Carballo-Mendivil et al. utilized only variables prior to enrollment and determined that XGBoost can provide high levels of sensitivity for identifying students who may be at-risk for dropping-out, while also providing the ability to identify interventions that can occur beginning with the first day of class [2]. Similarly, Cirak et al. created an early warning system for freshmen and identified that Random Forest provided better than Naive Bayes and Decision Tree models to predict if a freshman will be successful/unsuccessful after completing the first year [4].

Recent studies further support the utilization of ensemble methods in structured higher education prediction tasks. As previously discussed, Umendu et al. compared Logistic Regression, Random Forest, Gradient Boosting, and Extra Trees on the UCI dataset and found that Extra Trees provided the best overall results when implemented with nested cross-validation, feature optimization, SMOTE, and SHAP analysis [3]. Additionally, Bouihi et al. demonstrated that regularized regression models remain viable as simpler and more interpretable baseline models even though ensemble methods tend to provide better predictive accuracy [6]. Finally, Rabelo and Zarate proposed an ensemble model that utilized logistic regression, neural networks, and decision trees and reported significant dropout detection performance for institutions managing retention efforts

[7]. Overall, these studies indicate that structured tabular data represent a feasible and cost-effective basis for creating retention analytics projects; especially where computational resources are limited.

2.3 Learning Analytics, Behavioural Data, and Temporal Models

In addition to these two streams of research there has been considerable interest in using learning analytic data provided by Learning Management Systems (LMS) or Virtual Learning Environment (VLE). This body of research extends beyond simply analyzing static student records to analyze engagement behaviors, assessment behaviors, click-stream behavior and temporal learning behavior. For example, Chen and Cui have shown that time series behavior in LMS's can be used to predict early course performance [17]; Adnan et al. have compared how well different modeling approaches performed at varying percentiles of course length and concluded that, although the predictive power of the models improved as more data about course activity became available, this improvement was both analytically useful and highlighted a fundamental trade off between the accuracy of the models and the timing of interventions [16]; similarly, Qiu et al. have demonstrated that both behavioral and process based variables related to learning behavior can improve performance predictions in E-learning environments [27].

More recent publications continue to build upon this body of research. For example, Goren et al. studied early drop-out prediction across different academic disciplines and found that learning behavior contributed differently to predicting an early drop out depending on the discipline being studied and the point during the course when the prediction was made; however, as a general rule later academic achievements were better predictors than earlier data [22]. Similarly, Masood et al. presented a hybrid deep learning model that identified high risk students in VLE environments; and a systematic review of the existing literature by Alnasyan et al. indicated that models using Long Short Term Memory (LSTM), Convolutional Neural Networks (CNN)-Long Short Term Memory (CNN-LSTM), Deep Neural Networks (DNNs), etc., were commonly applied when the focus of analysis was to understand traceable behaviors over time, as opposed to understanding students through their static administrative records [8] [9]. The value of these studies is twofold. First, they demonstrate the potential of richer behavioral data to support more effective prediction of student outcomes. Second, however, these studies are inherently more complex in terms of the required preprocessing and feature engineering steps and the necessary temporal alignments that would likely exceed the capabilities of a single term capstone project.

2.4 Explainability and Decision-Support Orientation

A significant trend in current research has been away from purely predictive approaches toward the development of decision support tools (explainable) that provide some basis for understanding action(s). For example, Baranyi et al. have shown how a combination of the use of deep learning-based dropout prediction with SHAP and permutation importance may enhance the interpretability of dropout predictions [18]. In addition to evaluating the interpretability of machine learning methods, Islam et al. have explored multiple machine learning models using a variety of

explainability techniques (SHAP and LIME), developed an application-specific view of the explanations (a support-system perspective), and emphasized the importance of developing explanations that reflect both the transparency and trustworthiness of learning analytic models [1], especially where those models will be used directly with students and/or advisors.

The reason this shift is important is that dropout risk prediction is more than just a technical classification task. If a university identifies a student at-risk of dropping out they should know the specific reasons why that student was identified and what type of intervention would be plausible based on the specific risks the student is identified as facing. The use of text input in dropout prediction as a non-structured signal source has been demonstrated to offer utility over structured input alone for providing practical institutional support (versus competitive leaderboard performance) by Phan et al. [20]. Additionally, in a field experiment, Plak et al. [21] evaluated the effectiveness of an early warning system to identify and counsel students who were likely to drop out, thereby demonstrating that while predictive performance is critical, it is only one component in determining whether a model will facilitate earlier, better, and ultimately more effective interventions. In the context of this capstone, explainability will be considered not only as a support mechanism for model interpretation, but also as a way to compare whether important predictors remain stable across datasets.

2.5 Research Trends, Strengths, and Limitations

In all of the research studied, there were numerous similarities. Firstly, tree based ensemble modelling (Random Forest, XGBoost, Gradient Boosting and Extra Trees) have dominated the field in predicting student dropout or attrition from institutions using structured methodologies as per [1] through to [7] and [13]. Secondly, predictive model performance can be improved by incorporating Behavioural/Temporal Learning Analytics Features; however these require additional Data Engineering and may operate differently depending upon Discipline, Course or Time Point as indicated in [8], [9], [16], [17], [22]. Finally, Explainability is increasingly being expected in current research with varying degrees of implementation throughout the literature as per [1], [10], [18], [23], [24].

However, the literature also identifies significant areas where improvement is required. For example, many studies continue to place an emphasis on the overall Accuracy/AUC of a model rather than providing sufficient detail regarding Recall/Balanced Accuracy/Cost of Missing Students who are At Risk of Attrition as identified in [3], [15], [19]. Another area of weakness is related to Timing. A further limitation across much of the reviewed literature is that model performance is commonly validated within a single dataset or institution, leaving open the question of whether strong results, feature rankings, and explanations remain stable when the model is evaluated in a different educational environment. The majority of models that perform well are typically built upon First Semester or even Second Semester Student Academic Performance. While this does provide better Predictive Quality it significantly reduces the Window available for Intervention as outlined within [1], [3], [4], [7], [22].

2.6 Research Gap Analysis

Although there have been numerous research studies utilizing predictive models for education-related topics, very little of this has included cross-validation across datasets. Research studies provide excellent results for a specific educational environment such as an institution, a cohort, a course or even a specific dataset; however, few studies assess whether the logical approach used to develop the models will translate into other environments with greater differences. There are also issues concerning explanations. While there are several explainable AI studies that attempt to explain a model developed using data from a single source [1],[10],[18] much less is known about what explanatory factors remain consistent when changing from one set of data to another (i.e., one dataset, population, or learning environment). There is also a methodological issue. Most project-style studies use previous research as a starting point and continue developing upon that base without first duplicating the original study. Lastly, there continues to be an ongoing balance between predictive performance and timing of interventions. Later academic or behavioral variables may result in improved model performance while reducing the potential usefulness of an early warning system. The purpose of this capstone was to address these areas of need by first reproducing Islam et al. (2025) for their UCI dataset analysis. Next, the analysis was extended to OULAD as the main evaluation environment. Finally, the goal was to determine if both model performance and explanatory pattern remained consistent after the features and labels were harmonized.

3. Data Source Identification

A two-dataset approach has been adopted to make use of two publically available datasets for the complimentary objectives of evaluation and reproducibility/benchmarking. As the main evaluation data source OULAD is used due to its rich behavioral and temporal characteristics providing a robust setting for testing model performance. In addition to evaluating our proposed solution with this comprehensive and realistic learning analytic environment, we also plan to use the UCI "Predict Students' Dropout and Academic Success" as the reproduction and benchmark dataset of the base paper. The adoption of a two-dataset approach allows us to contribute to the methodology by being able to perform a cross-dataset generalization analysis.

3.1 Dataset Summary

The updated data strategy employs an additional publicly available education dataset that serves as a secondary role. The OULAD was selected as the main or primary data set because it provides the richest behaviorally based learning data that includes temporal elements. A second publicly available education dataset, referred to herein as the UCI dataset will be employed in order to replicate results of the original research and provide a controlled comparative structure for evaluating each model.

Dataset	Role	Format	Size	Structure	Target / Outcome	Access	License / Use
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UCI Predict Students' Dropout and Academic Success	Primary dataset	CSV and ZIP download	520.8 KB ZIP; about 496 KB CSV	4,424 rows; 37 columns including Target (36 predictors + 1 target)	Dropout / Enrolled / Graduate	Direct public download from UCI, no login required	Public repository access with dataset DOI and citation requirement [25]
Open University Learning Analytics Dataset (OULAD)	extension dataset	ZIP containing multiple CSV tables	44.6 MB ZIP download	22 module-presentations; 32,593 students; multiple linked tables; 10,655,280 click entries in studentVle logs	final_result, withdrawal-related outcome patterns, assessment performance, VLE engagement	Direct public download from the OU Analyse website [26]	CC BY 4.0 [26]

3.2 Primary Evaluation Dataset Rationale

A major difference in this proposed version of the project from the previous is that we will use the Open University Learning Analytics Dataset (OULAD) instead of the previously stated UCI benchmark to evaluate our model's performance. This is particularly important because OULAD offers much more detailed behavioral and temporal data regarding student learning behavior than does the UCI benchmark; consequently, we can better assess how well our models are able to predict results when they have been trained on and tested with rich data. When I inspected the official archive containing these datasets, I found there were 7 CSVs in total that included studentInfo, studentRegistration, studentAssessment, and studentVLE. There were 32,593 records for each student record in the studentInfo dataset. The final result field contained 4 possible values (Distinction, Pass, Fail, or Withdrawn). In addition to those records, there were 173,912 assessments taken by students and over 10 million interactions within the VLE. Because of its size relative to the smaller UCI benchmark dataset, using OULAD to train/test a model that predicts at-risk students would represent a more realistic approach. Therefore, this change addresses the criticism of our original proposal being overly incremental.

3.3 Reproduction and Benchmark Dataset Rationale

The UCI Predict Students' Dropout and Academic Success Dataset was selected for retention and is being utilized with greater specificity than originally planned. This data will function as the Reproduction/Benchmark Data Set for the Base Paper [1],[25].

A review of the data available on the internet as a CSV file confirmed that there were 4,424 entries in the file and 37 columns. All 37 columns were used to create 36 Predictor Variables and a single Target Variable labeled "Target". The original Target Categories are Graduate, Dropout, and Enrolled.

This data is useful in this project for several reasons. First and foremost, the use of this dataset enables the project to replicate the primary workflow used by Islam et al. (2025), but at significantly less cost and complexity than if an attempt had been made to extend or expand upon

their work prior to replicating theirs. Second, the OULAD is a much more behaviorally rich data set; therefore, this dataset will provide a structured reference point to compare it against.

3.4 Preliminary Data Assessment

Beginning with the inspection of the original data set prior to performing any preprocessing, it was confirmed that this particular data set was appropriate for model building due to the fact that it is small in size, well structured and could be easily run within a typical laptop or Google Colab environment.

- There was a total of 4424 students represented in the UCI downloaded file and there were a total of 37 columns of data. Of these, 36 columns represented predictor variables while the remaining single column was the target variable referred to as "Target".
- As noted earlier, the three target categories are Graduate (2209 records / 49.93%); Dropout (1421 records / 32.12%); and Enrolled (794 records / 17.95%) which indicates a moderate level of class imbalance however this does not indicate an extreme imbalance.
- The preliminary audit showed no missing values and no duplicate rows in the publicly available .csv file used here.
- The predictor variables contained in all of the columns appear to be numeric encoded; however, many of them are clearly representing categorical or ordinal concepts such as course, gender, whether or not they have a scholarship, their nationality, their parents' educational history, etc., and/or semester unit measures. These will need to be carefully handled in terms of data types rather than being treated as if they were simply continuous values.
- Because there are both first- and second-semester academic performance variables included in the data set this should provide very strong predictive capabilities. However, this may limit how quickly the warning system actually provides warnings. This time factor is analytically significant and will be addressed directly when developing models.
- A preliminary assessment further indicates that OULAD will be possible but significantly more difficult. The archive has seven interrelated tables and the biggest table in the archive (student VLE) has over 10 million individual interactions stored within it. Therefore, the major obstacle is not gaining access to the data itself, but rather designing pre-processing logic: table joins, temporal aggregations on VLE behavior, and carefully controlling potential "leakage". Thus, due to these challenges, we believe that OULAD would provide a good source from which to study both transferability and robustness using OULAD as our primary evaluation dataset; however, this will likely result in the need for greater care when developing features compared to UCI.

4. Preliminary Analytical Framework

This framework was developed in an effort to create a realistic, actionable, and comparable model of both current research and the constraints that the project has. This will occur as follows.

4.1 Planned Workflow

This modified version of the structure is created for response to the base document as well as the professor's comments. The analytical process flow will be divided into four phases.

- Phase 1 - Replication of Explainable Machine Learning Workflow at UCI:
The project will replicate the fundamental workflow described by Islam et al. (2025), concerning the use of explainable machine learning, on the UCI data set. This initial replication phase will ensure that there is a valid baseline for comparison with any extensions implemented during subsequent phases.
- Phase 2 - Creation of an OULAD Student-Risk Prediction Workflow:
Next, the project will develop a similar student-risk prediction workflow for students in the OULAD system, utilizing demographic, registration, assessments, and early behaviour variables. The primary focus in this second phase will be to identify students who withdraw from their studies vs. those who do not withdraw.
- Phase 3 - Evaluation of Cross- Dataset Generalization:
Once each of the two data sets has been validated using a defensible pipeline, the project will evaluate how effectively the pipelines can generalize across different data sets using a common binary attrition representation. For example, in the case of UCI dropout represents attrition and enrolled plus graduate represents persistence. In contrast, withdrawal represents attrition while pass, fail, and distinction represent persistence in the OULAD dataset.
- Phase 4 - Comparative Explanation Stability Analysis:
Finally, the fourth analytical phase will compare feature importance rankings and SHAP explanation patterns across the two data sets in order to understand which factors or predictors remain consistent or "stable" across educational settings, and which are unique to one particular dataset.

4.2 Evaluation and Validation Strategy

The evaluation method for this student-risk project will look at measures other than just the overall percentage of correct predictions as a way to evaluate a student risk project since you cannot judge the effectiveness of a student-risk project based solely on how many students are correctly identified as being at risk. Macro F1-score, dropout-class recall, balanced accuracy, Multi-class ROC-AUC will be the primary metrics for comparing models. The confusion matrix will be used to demonstrate if the model is incorrectly identifying an enrolled or Graduate student as a dropout which could limit the usefulness of interventions.

Validation will start using stratified train/test splitting and cross validation. After validation, the top performing models will undergo more fine-tuned tuning and error analysis.

In addition to within-dataset validation, cross-dataset testing will serve as a form of external validation. A key part of the analysis will be the comparison between internal model performance

and the level of performance degradation observed when a model is transferred to the other dataset. This will provide direct evidence regarding generalization and robustness.

4.3 Why These Methods Fit the Project

The chosen methodologies were designed to address the business issue at hand as well as the features of the provided data. The UCI data set is formatted such that it is structured, tabular and moderate in size which allows for a variety of different types of classical or ensemble-based machine learning applications but may be too small an environment to implement heavy deep learning infrastructure. An advantage of using logistic regression is the clarity of the baseline model. A benefit of decision trees is their ease of interpretation. Random Forest and XG Boost have been shown in prior studies (i.e., [1] through [9], [13]) to provide high quality results on educational classification problems; the SHAP methodology will help to explain why these two models are providing better results than the others. Not only are these methods successful on structured education data, they are useful in that they provide a means of fairly comparing how well reproduction and extension stages work.

These methods are suitable not only because they are effective on structured educational data, but also because they allow a fair comparison between reproduction and extension stages. In the revised project, SHAP is not used only to explain a single best-performing model, but also to compare whether explanation patterns remain stable across datasets

5. Feasibility and Constraints Assessment

The project still fits into an academic term, but it has improved methodologically compared to the previous version. Both the UCI Benchmark and OULAD are publically available, so they can all be run using the same python libraries (Pandas, Scikit-Learn, XGBoost & SHAP). The UCI Benchmark is small and easy to use right away, while OULAD is large due to the nature of the data collected for interaction logs and not because of any specialized hardware or software requirements. Therefore, the primary limitations are going to be methodological in nature and not just based on technology. To work with OULAD you need to join tables, create time-based features and implement methods that limit how much data you have access to during training/evaluation. Additionally, cross-dataset experiments will require mapping labels from each dataset into some common set of labels and selecting a suitable subset of features across datasets. In order to keep this project at a reasonable size we plan to focus our efforts on reproducing results, evaluating robustness, and creating explanations as opposed to producing polished dashboards or full-scale production deployments.

Finally, there are constraints related to ethics of interpretation. Although both datasets are anonymized and publicly available [25][26] the ultimate goal of this project is to create a decision-support tool and not an automated decision-making tool. Student attrition is influenced by a variety of academic, financial and psychosocial factors [12]. As such, the predictions produced through this model should be used to inform how best to allocate resources for interventions rather than to

classify students deterministically. These limitations are also relevant to discussions surrounding business value as they pertain to whether the proposed model will provide adequate support for interventions or will need to supplant human judgment.

Overall, the Capstone can be completed within one academic term by utilizing publically available datasets with a specific and limited number of machine learning models as well as a transparent and explainable workflow; therefore, completion does not depend on having fully developed products. The added benefit in the project is now through reproducibility across datasets, explanation stability, and cross dataset comparison rather than the complexity of interfaces.

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